Micro-Service Hub Platform

Test Case Design

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1.Real-Time Chat

Field	Details
Test Case ID	TC-MSH-EP-BVA-01
Test Scenario	Verify that the system accepts valid chat messages and rejects empty or over-limit messages.
Preconditions	1. User account is active and logged in. 2. User has an active chat with another user. 3. Maximum message length is 1000 characters.
Test Steps	1. Open the active chat. 2. Enter a message with the specified length. 3. Click Send. 4. Observe whether the message is accepted or rejected.
Input Data	**Equivalence Partitioning (EP):** • EP-CHAT-01: Valid message → Accept/Send • EP-CHAT-02: Empty message → Reject • EP-CHAT-03: More than 1000 characters → Reject **Boundary Value Analysis (BVA):** • BVA-CHAT-01: 0 characters → Reject • BVA-CHAT-02: 1 character → Accept • BVA-CHAT-03: 999 characters → Accept • BVA-CHAT-04: 1000 characters → Accept • BVA-CHAT-05: 1001 characters → Reject
Expected Result	System accepts valid messages within the 1–1000 character limit and rejects empty messages and messages exceeding 1000 characters.
Actual Result	System accepted valid messages and rejected empty and over-limit messages.
Status	Pass
Remarks	Validates message input using Equivalence Partitioning (EP) and Boundary Value Analysis (BVA).

2.Online Payment

Field	Details
Test Case ID	TC-MSH-EP-BVA-02
Test Scenario	Verify that the system accepts valid payment amounts and rejects amounts outside the allowed range or non-numeric input.
Preconditions	1. User account is active and logged in. 2. A service has been selected for payment. 3. Allowed payment range is ■100–■1,000.
Test Steps	1. Open the payment page. 2. Enter a payment amount. 3. Click Pay. 4. Observe whether the payment is accepted or rejected.
Input Data	**Equivalence Partitioning (EP):** • EP-PAY-01: ■50 → Below minimum → Reject • EP-PAY-02: ■500 → Valid → Accept • EP-PAY-03: ■1,000 → Valid → Accept • EP-PAY-04: ■1,100 → Above maximum → Reject • EP-PAY-05: ABC → Non-numeric → Reject **Boundary Value Analysis (BVA):** • BVA-PAY-01: ■99 → Reject • BVA-PAY-02: ■100 → Accept • BVA-PAY-03: ■101 → Accept • BVA-PAY-04: ■999 → Accept • BVA-PAY-05: ■1,000 → Accept • BVA-PAY-06: ■1,001 → Reject
Expected Result	System accepts payment amounts from ■100 to ■1,000 and rejects amounts below ■100, above ■1,000, or non-numeric input.
Actual Result	System accepted valid payment amounts and rejected invalid amounts and non-numeric input.
Status	Pass
Remarks	Validates payment amount using Equivalence Partitioning (EP) and Boundary Value Analysis (BVA).

3.Secure Payment Hold & Release

Field	Details
Test Case ID	TC-MSH-EP-03
Test Scenario	Verify that the system correctly holds or releases payment based on the service/payment status.
Preconditions	1. A service has an associated payment. 2. The service is in a valid workflow state. 3. Payment hold and release rules are enabled.
Test Steps	1. Open the service/payment status page. 2. Check the current service status. 3. Observe whether payment is held or released according to the status.
Input Data	**Equivalence Partitioning (EP):** • EP-HOLD-01: IN_PROGRESS → Service not completed → Payment Hold • EP-HOLD-02: CONFIRMED → Completed & confirmed → Payment Release • EP-HOLD-03: DISPUTED → Disputed → Payment Hold
Expected Result	System keeps payment on hold for IN_PROGRESS and DISPUTED states and releases payment for the CONFIRMED state.
Actual Result	System held payment for IN_PROGRESS and DISPUTED states and released payment for the CONFIRMED state.
Status	Pass
Remarks	Validates the status partitions using the Equivalence Partitioning (EP) technique.